

antivirus program created by Ray Tomlinson in 1972 and was capable of deleting the Creeper virus.

Antivirus software works mainly using signature and behavioural based techniques. In signature based detection, when any new malware is discovered by researchers, its file signature is extracted and added to the database of the antivirus. During a security scan, the antivirus checks for similar signatures in the files it scans, and if any match is found then it's detected as malware. In behavioural detection, the algorithm or the intention of a malicious program is understood and all suspicious behaviour is added to the database of the antivirus. During the execution of a file, the antivirus monitors the executed program and tries to find behaviour that matches with the database. If any match is found, it's detected as malware. Besides these two techniques, other methods like sandbox detection, data mining techniques, heuristic analysis, rootkit detection and real-time protection are also used.

Sandboxing uses a virtual environment to run the suspected program and isolate it from the real environment. It then checks the behaviour of the program and compares it with the database; if any match is found, it is flagged as malware. This method is safe and protects the system from any malware attack.

The data mining technique uses machine learning algorithms to classify malware based on its behaviour.

Heuristic analysis is a technique used to identify the genre of a virus or malware. Usually, attackers can create variants of a virus and there can be a family of similar viruses, forming a genre. It is easier to identify the signature of the genre than to detect a specific virus. Heuristic analysis creates the signature genre of various families of viruses based on their behaviour, which is stored in the database of the antivirus. When the behaviour of a program matches the genre of a virus, it is flagged as a virus belonging to that family.

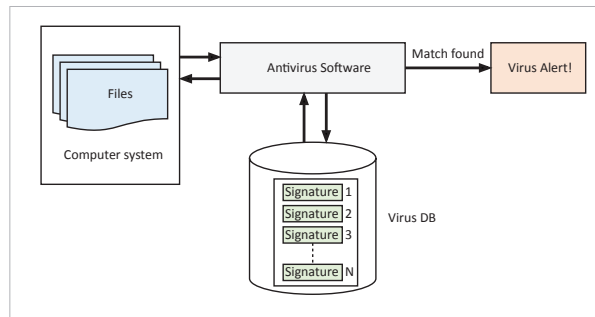


Figure 2: Virus detection process

Rootkit detection is used particularly for malware called rootkit, which can take administrative control over the computer. A rootkit can tamper with the antivirus, disabling it, and is also sometimes very difficult to remove. A real-time antivirus scans each and every process executed in the background in real time. This protection is very important and prevents unwanted malware from being activated.

Whatever technique is used by the antivirus software, the most challenging thing is the prevention of any false positive detection. Many programs may behave like malware or carry a file signature that looks like malware, but never do any harm. But many antivirus or anti-malware software sometimes detect such programs as malware.

For example, here is a C program using the `printf()` function:

```
#include <stdio.h>
main()
{
    int n;
    n=1;
    while (n>0)
    {
        printf("Enter a number: ");
        scanf("%d",&n);
        printf("The number you entered is: %d\n", n);
    }
}
```

The above code will be detected as malware by some antivirus when

compiled. But when that code is written using only the `puts()` function instead of `printf()`, it is not detected as malware.

The following code is written by replacing `printf()` with `puts()`:

```
#include <stdio.h>
```

```
main()
{
    int n;
    n=1;
    while (n>0)
    {
        puts("Enter a number: ");
        scanf("%d",&n);
        puts("The number you entered is: %d\n", n);
    }
}
```

The above code can be compiled and run without any malware detection. This is because `printf()` has uncontrolled format string security risk, and it's possible to carry a malicious act of printing data from the call stack or possibly other locations in memory using format tokens like `%s` and `%x`. One can also command `printf()` to write arbitrary data to any location using the `%n` format token, and also write the number of bytes formatted to an address stored on the stack.

Even though ClamAV is a powerful protection engine and uses various techniques for malware detection, the false positive is always a concern for the user. However, some antivirus testing companies have tested and found it to be good. The Shadowserver Foundation (a non-profit security organisation) tested ClamAV for six months in 2011 and found that it was able to detect 75.45 per cent of viruses, putting it in fifth place behind AhnLab, Avira, BitDefender and Avast. AhnLab, the top antivirus, detected